Computing Laboratory · Fakultas Informatika · Telkom University

Advance Study Group

AI, ML, dan Optimasi

Pertemuan 0



Agenda Hari Ini

0:00	Kenapa AI? Cerita dari paper (8 menit)
0:08	Intuisi: AI, ML, dan optimasi (27)
0:35	Capaian dan harapan kelas (20)
0:55	Istirahat (5)
1:00	Stasiun persiapan (10)
1:10	Flappy Komodo: manusia lawan evolusi (45)
1:55	Pemenang dan Pertemuan 1 (5)

AI yang kamu pakai bermula dari paper



Contents lists available at ScienceDirect

Applied Soft Computing

journal homepage: www.elsevier.com/locate/asoc

Komodo Mlipir Algorithm

Suyanto Suyanto *, Alifya Aisyah Ariyanto, Alifya Fatimah Ariyanto

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ARTICLE INFO

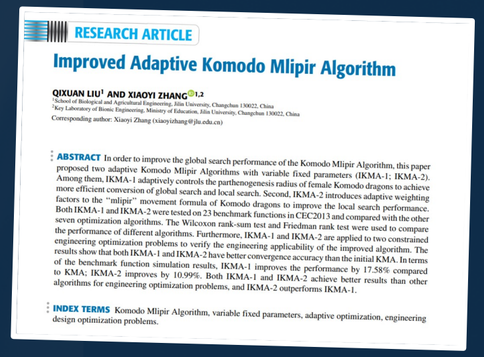
Article history:
Received 15 March 2021
Received in revised form 18 October 2021
Accepted 4 November 2021
Available online 17 November 2021

Keywords:
Komodo mlipir algorithm
Metaheuristic optimization
Self-adaptation of population
Exploitation–exploration balance
Scalable to thousand dimensions

ABSTRACT

This paper proposes Komodo Mlipir Algorithm (KMA) as a new metaheuristic optimizer. It is inspired by two phenomena: the behavior of Komodo dragons living in the East Nusa Tenggara, Indonesia, and the Javanese gait named mlipir. Adopted the foraging and reproduction of Komodo dragons, the population of a few Komodo individuals (candidate solutions) in KMA are split into three groups based on their qualities: big males, female, and small males. First, the high-quality big males do a novel movement called high-exploitation low-exploration to produce better solutions. Next, the middle-quality female generates a better solution by either mating the highest-quality big male (exploitation) or doing parthenogenesis (exploration). Finally, the low-quality small males diversify candidate solutions using a novel movement called mlipir (a Javanese term defined as a walk on the side of the road to reach a particular destination safely), which is implemented by following the big males in a part of their dimensions. A self-adaptation of the population is also proposed to control the exploitation–exploration balance. An examination using the well-documented twenty-three benchmark functions shows that KMA outperforms the recent metaheuristic algorithms. Besides, it provides high scalability to optimize thousand-dimensional functions. The source code of KMA is publicly available at: https://suyanto.staff.telkomuniversity.ac.id/komodo-mlipir-algorithm and https://www.mathworks.com/matlabcentral/fileexchange/102514-komodo-mlipir-algorithm.

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RESEARCH ARTICLE

Improved Adaptive Komodo Mlipir Algorithm

QIXUAN LIU[1] AND XIAOYI ZHANG[1,2]

[1]School of Biological and Agricultural Engineering, Jilin University, Changchun 130022, China
[2]Key Laboratory of Bionic Engineering, Ministry of Education, Jilin University, Changchun 130022, China
Corresponding author: Xiaoyi Zhang (xiaoyizhang@jlu.edu.cn)

ABSTRACT In order to improve the global search performance of the Komodo Mlipir Algorithm, this paper proposed two adaptive Komodo Mlipir Algorithms with variable fixed parameters (IKMA-1; IKMA-2). Among them, IKMA-1 adaptively controls the parthenogenesis radius of female Komodo dragons to achieve more efficient conversion of global search and local search. Second, IKMA-2 introduces adaptive weighting factors to the "mlipir" movement formula of Komodo dragons to improve the local search performance. Both IKMA-1 and IKMA-2 were tested on 23 benchmark functions in CEC2013 and compared with the other seven optimization algorithms. The Wilcoxon rank-sum test and Friedman rank test were used to compare the performance of different algorithms. Furthermore, IKMA-1 and IKMA-2 are applied to two constrained engineering optimization problems to verify the engineering applicability of the improved algorithm. The results show that both IKMA-1 and IKMA-2 have better convergence accuracy than the initial KMA. In terms of the benchmark function simulation results, IKMA-1 improves the performance by 17.58% compared to KMA; IKMA-2 improves by 10.99%. Both IKMA-1 and IKMA-2 achieve better results than other algorithms for engineering optimization problems, and IKMA-2 outperforms IKMA-1.

INDEX TERMS Komodo Mlipir Algorithm, variable fixed parameters, adaptive optimization, engineering design optimization problems.



An Empirical Analysis of Skin-Tone Bias and the Limitations of Random Oversampling in Dermatology AI

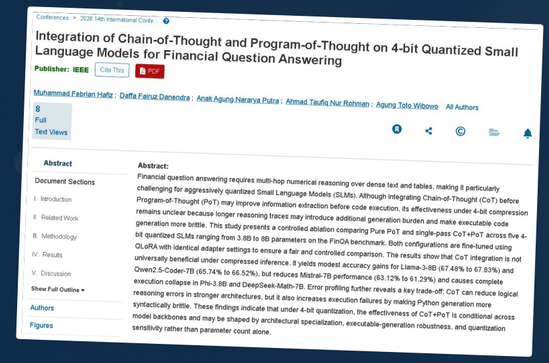
Publisher: IEEE Cite This PDF

Ahmad Daffa Aminuddin Siregar ; Kemas Muslim Lhaksmana ; Andy Maulana Yusuf All Authors

7 Full Text Views

Abstract
Document Sections
I. Introduction
II. Related Work
III. Methodology
IV. Experiments
V. Results
Show Full Outline
Authors

Abstract:
Convolutional neural networks (CNNs) are highly effective at detecting skin cancer, but their aggregate accuracy can conceal performance disparities across patient demographics. This is concerning for widely used corpora such as HAM10000, in which the overwhelming majority of images depict light skin. We audit a standard EfficientNet-B0 skin-cancer classifier for skin-tone bias and test whether data-level mitigation can close any observed gap. Using a leakage-free pipeline evaluated over three random seeds, we measure the Equal Opportunity Difference (EOD) between light- and dark-skinned groups, and evaluate two interventions: random oversampling and skin-tone sample weighting. We find the baseline recall gap is modest (EOD approximately 0.11) and, critically, that its 95% bootstrap confidence interval spans zero. The same holds after both interventions: neither produces a statistically distinguishable change. Our central finding is methodological: HAM10000 contains only about 182 dark-skin images (roughly 1.8% of the corpus), leaving only about 10 dark-skin malignant cases in the test set. With a recall granularity of 0.10 per patient and a confidence-interval width near 0.65, fairness cannot be measured reliably from this dataset, and duplicating 152 images adds no new information. A quantitative Grad-CAM border analysis nonetheless reveals a consistent directional trend: both interventions increase the model's attention to image borders on dark skin. We conclude that naive data-level fixes are not a substitute for the prospective collection of genuinely diverse, expert-labeled dermatological data.



Conferences > 2026 14th International Confe...

Integration of Chain-of-Thought and Program-of-Thought on 4-bit Quantized Small Language Models for Financial Question Answering

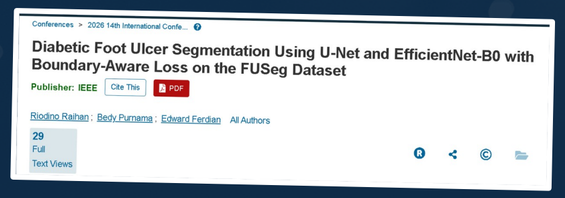
Publisher: IEEE Cite This PDF

Muhammad Febrian Hafiz ; Daffa Fairuz Danendra ; Anak Agung Narayya Putra ; Ahmad Taufiq Nur Rohman ; Agung Toto Wibowo All Authors

8 Full Text Views

Abstract
Document Sections
I. Introduction
II. Related Work
III. Methodology
IV. Results
V. Discussion
Show Full Outline
Authors
Figures

Abstract:
Financial question answering requires multi-hop numerical reasoning over dense text and tables, making it particularly challenging for aggressively quantized Small Language Models (SLMs). Although integrating Chain-of-Thought (CoT) before Program-of-Thought (PoT) may improve information extraction before code execution, its effectiveness under 4-bit compression remains unclear because longer reasoning traces may introduce additional generation burden and make executable code generation more brittle. This study presents a controlled ablation comparing Pure PoT and single-pass CoT+PoT across five 4-bit quantized SLMs ranging from 3.8B to 8B parameters on the FinQA benchmark. Both configurations are fine-tuned using QLoRA with identical adapter settings to ensure a fair and controlled comparison. The results show that CoT integration is not universally beneficial under compressed inference. It yields modest accuracy gains for Llama-3-8B (67.48% to 67.83%) and Qwen2.5-Coder-7B (65.74% to 66.52%), but reduces Mistral-7B performance (63.12% to 61.29%) and causes complete execution collapse in Phi-3.8B and DeepSeek-Math-7B. Error profiling further reveals a key trade-off: CoT can reduce logical reasoning errors in stronger architectures, but it also increases execution failures by making Python generation more syntactically brittle. These findings indicate that under 4-bit quantization, the effectiveness of CoT+PoT is conditional across model backbones and may be shaped by architectural specialization, executable-generation robustness, and quantization sensitivity rather than parameter count alone.



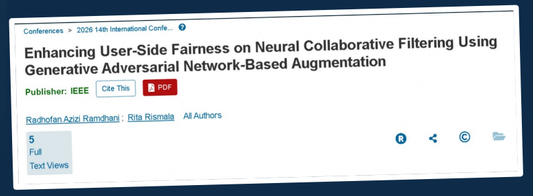
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Diabetic Foot Ulcer Segmentation Using U-Net and EfficientNet-B0 with Boundary-Aware Loss on the FUSeg Dataset

Publisher: IEEE Cite This PDF

Riodino Raihan ; Bedy Purnama ; Edward Ferdian All Authors

29 Full Text Views



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Enhancing User-Side Fairness on Neural Collaborative Filtering Using Generative Adversarial Network-Based Augmentation

Publisher: IEEE Cite This PDF

Radhofan Azizi Ramdhani ; Rita Rismala All Authors

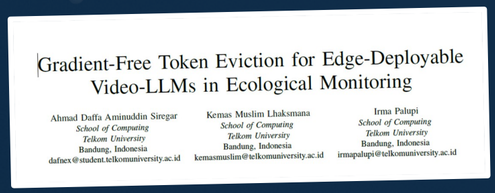
5 Full Text Views



Reference-vector competitor pairing for constrained large-scale multi-objective competitive swarm optimisation

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[a]School of Computing, Telkom University, Bandung, Indonesia
[b]Computing Laboratory, Telkom University, Bandung, Indonesia
[c]Center of Excellence of Artificial Intelligence for Learning and Optimization, Telkom University, Bandung, Indonesia



Gradient-Free Token Eviction for Edge-Deployable Video-LLMs in Ecological Monitoring

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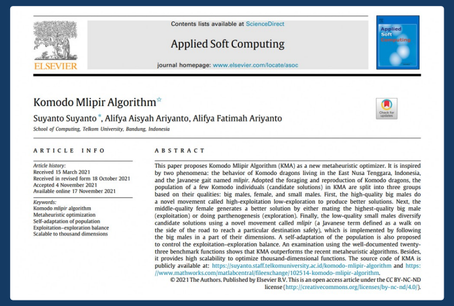
Irma Palupi
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Banyak yang ditulis di kampus kita sendiri

CERITA

Satu ide dari kampus kita, tiga paper

2021



Contents lists available at ScienceDirect

Applied Soft Computing

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Komodo Mlipir Algorithm☆

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ARTICLE INFO

Article history:
Received 15 March 2021
Received in revised form 18 October 2021
Accepted 4 November 2021
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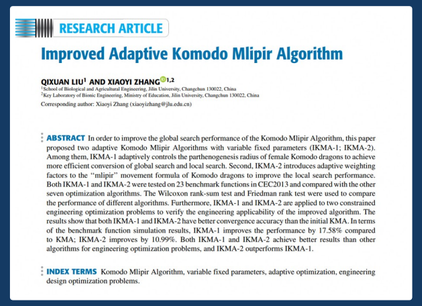
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Komodo Mlipir Algorithm

Ide baru, Telkom University

2022



RESEARCH ARTICLE

Improved Adaptive Komodo Mlipir Algorithm

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[1]School of Biological and Agricultural Engineering, Jilin University, Changchun 130022, China
[2]Key Laboratory of Bionic Engineering, Ministry of Education, Jilin University, Changchun 130022, China
Corresponding author: Xiaoyi Zhang (xiaoyizhang@jlu.edu.cn)

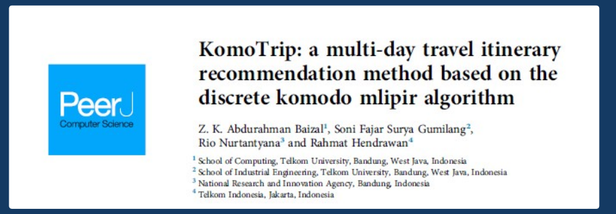
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INDEX TERMS Komodo Mlipir Algorithm, variable fixed parameters, adaptive optimization, engineering design optimization problems.

Diperbaiki orang lain

Jilin University, Tiongkok

2025



PeerJ Computer Science

KomoTrip: a multi-day travel itinerary recommendation method based on the discrete komodo mlipir algorithm

Z. K. Abdurahman Baizal[1], Soni Fajar Surya Gumilang[2], Rio Nurtantyana[3] and Rahmat Hendrawan[4]
[1] School of Computing, Telkom University, Bandung, West Java, Indonesia
[2] School of Industrial Engineering, Telkom University, Bandung, West Java, Indonesia
[3] National Research and Innovation Agency, Bandung, Indonesia
[4] Telkom Indonesia, Jakarta, Indonesia

Dipakai: KomoTrip

Menyusun rute wisata beberapa hari

Tiga Kata, Satu Peta

AI

Payung besarnya

Komputer mengerjakan tugas yang biasanya butuh kecerdasan.

Catur, terjemahan, self-driving car

Machine Learning

Bagian dari AI

Aturannya dipelajari dari contoh, tidak ditulis tangan.

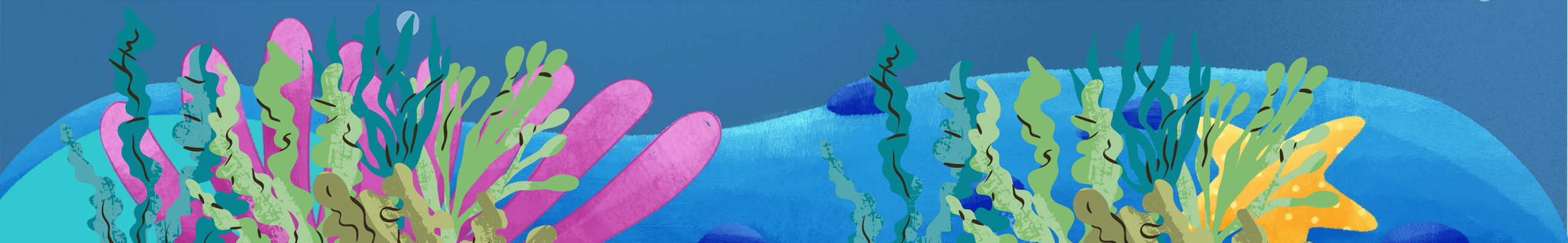
Filter spam, rekomendasi video, pengenal wajah

Optimasi

Mesin di balik keduanya

Mencari pilihan terbaik dari sangat banyak kemungkinan.

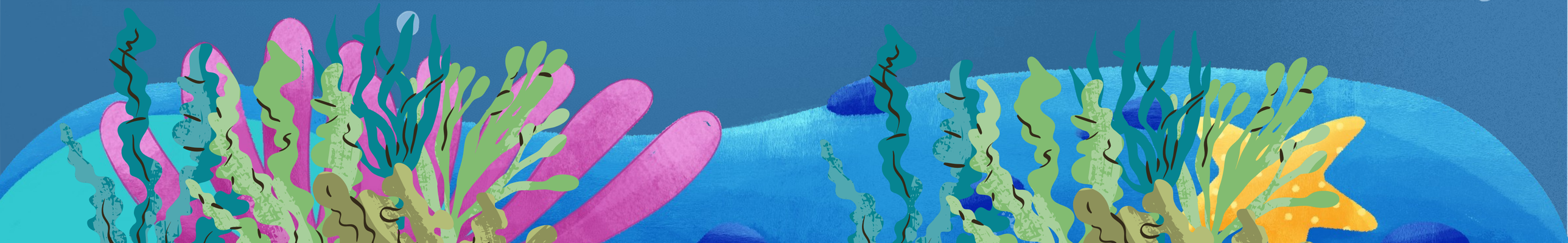
Rute pengiriman, jadwal kuliah, melatih model



Menulis Aturan atau Belajar Aturan

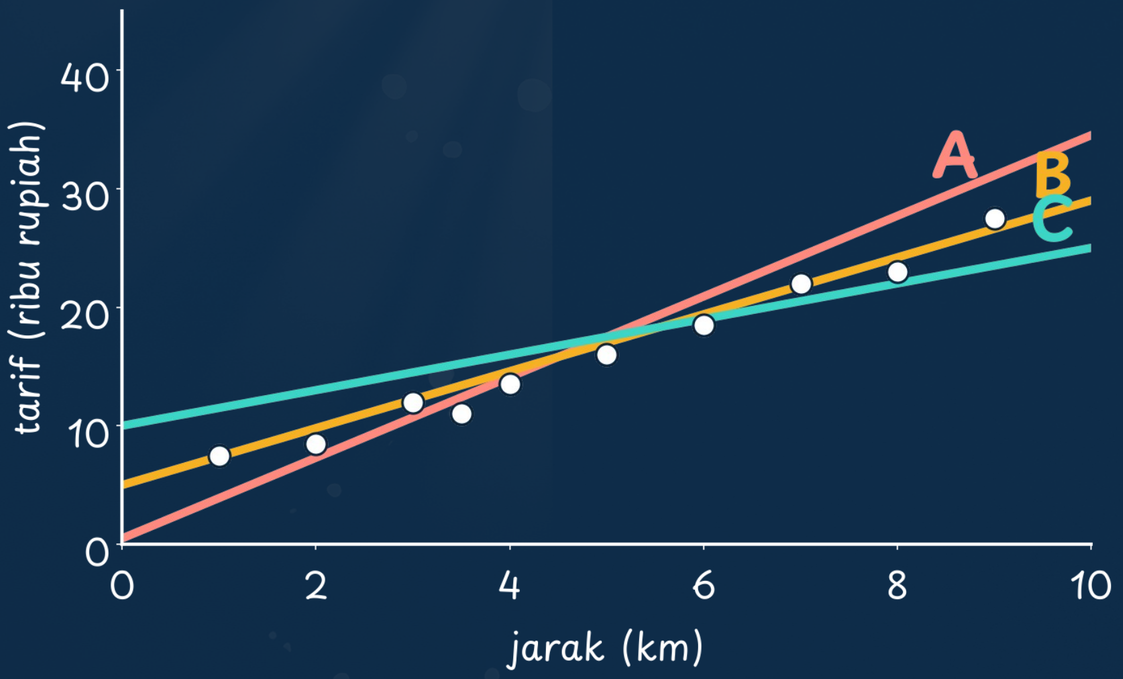
Program biasa	**Aturan**	**Data**	→	**Jawaban**
Machine learning	**Data**	**Jawaban**	→	**Aturan**

- Di Go kamu menulis: if jarak > 5 { tarif += ... }
- Di ML kamu memberi ribuan contoh perjalanan, lalu komputer mencari sendiri rumus tarifnya



AKTIVITAS

Garis mana yang paling cocok?



Data

10 perjalanan ojek: jarak dan tarif

Model

tarif = a × jarak + b

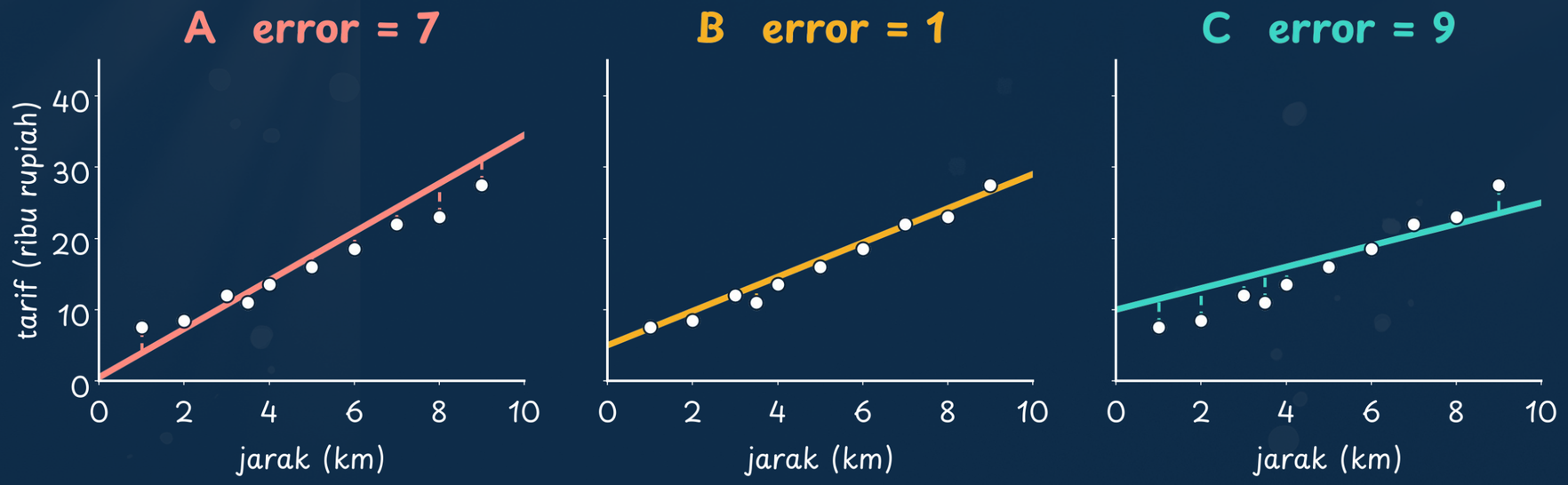
a dan b adalah kenopnya

Angkat tangan

A, B, atau C? Apa alasanmu?

AKTIVITAS

Ubah "kelihatannya pas" menjadi angka

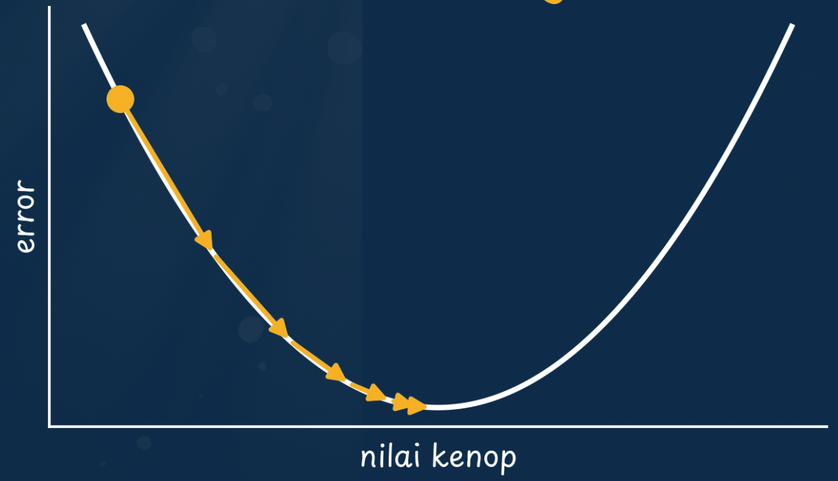


Error = rata-rata kuadrat selisih. **Belajar = memutar kenop sampai error terkecil.**

INTUISI

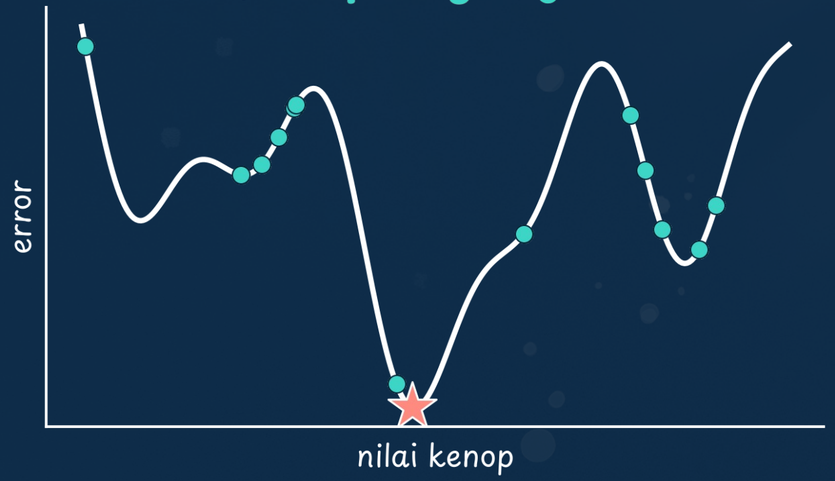
Dua cara memutar kenop

Ikuti kemiringan



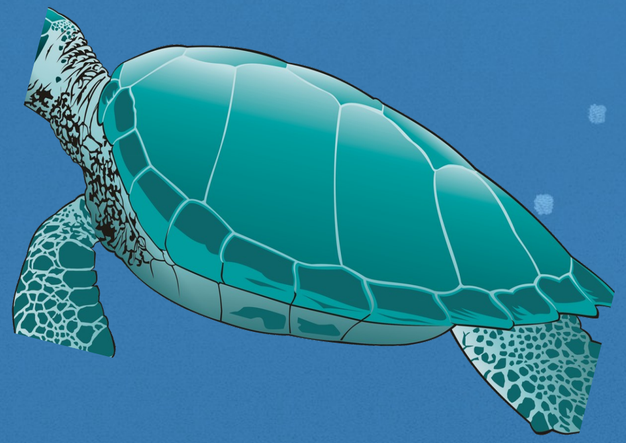
Gradient descent. Pekan 2 sampai 7, buku UDL

Coba, simpan yang terbaik



Metaheuristik. Pekan 9 sampai 12, buku Luke

Evolusi Gambar



Satu aturan

Ubah sedikit. Simpan jika lebih mirip target, buang jika tidak.



Perhatikan

Kapan kurva turun cepat? Kapan melambat?

Empat Kata yang Muncul Setiap Pekan

Model

Rumus dengan kenop

tarif = a × jarak + b

Error

Seberapa salah modelnya

Aplikasi bilang tiba 10 menit, nyatanya 25

Pencarian

Cara memutar kenop

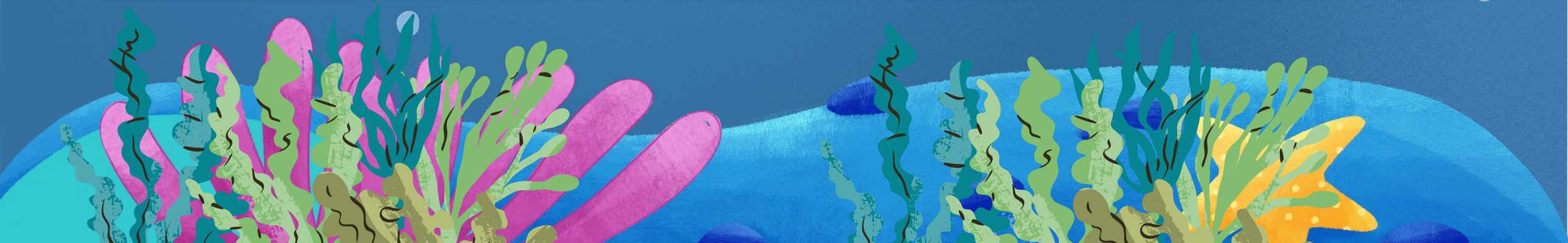
Tambah gula sedikit demi sedikit sampai kopinya pas

Anggaran

Berapa kali boleh mencoba

Hanya tiga kali cicip sebelum tamu datang

Keempatnya akan kamu pakai di game penutup hari ini



Di Akhir Pekan 16

- Melatih jaringan saraf yang kamu tulis sendiri, lalu dengan PyTorch
- Menjelaskan mengapa model belajar: error, gradien, backpropagation
- Menilai model dengan jujur: data uji, kebocoran data
- Menulis dan membandingkan metaheuristik: SA, GA, PSO
- **Proyek tim dengan demo yang bisa dibuka orang lain**

Peta 16 Pekan

Pekan 1 sampai 7

Dasar ML

Error, gradient descent, klasifikasi, jaringan saraf, backprop, PyTorch, evaluasi jujur

Pekan 8

Checkpoint 1

Cek lisan individu. Tim proyek dibentuk

Pekan 9 sampai 12

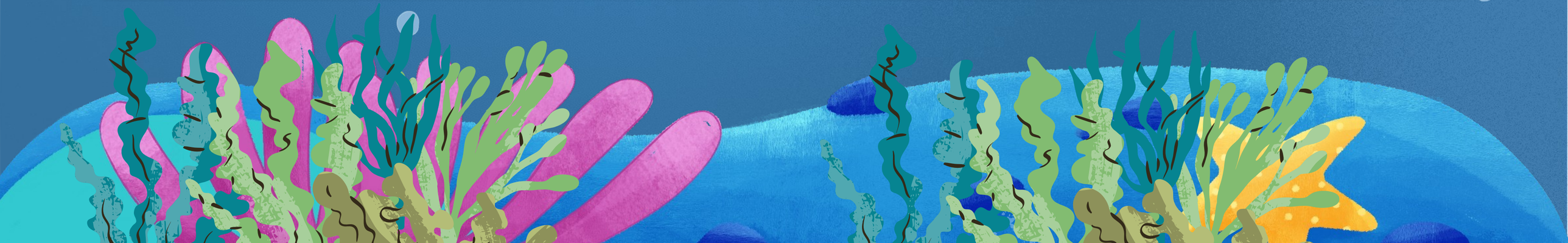
Metaheuristik

Simulated annealing, GA, PSO, representasi, optimasi di dalam ML

Pekan 13 sampai 16

Modern dan proyek

Transfer learning, Transformer dan LLM, deploy, Hari Demo



Ritme Mingguan

Sekitar 5 sampai 6 jam per pekan

1. Baca

Bab buku dengan panduan baca berbahasa Indonesia.

2. Kelas

2 jam: kuis pembuka tanpa perangkat, contoh, latihan berpasangan.

3. Latihan

Notebook Colab dan satu pertanyaan pemahaman.



Dua Buku, Gratis dan Sah

Understanding Deep Learning

Simon J. D. Prince, MIT Press 2023

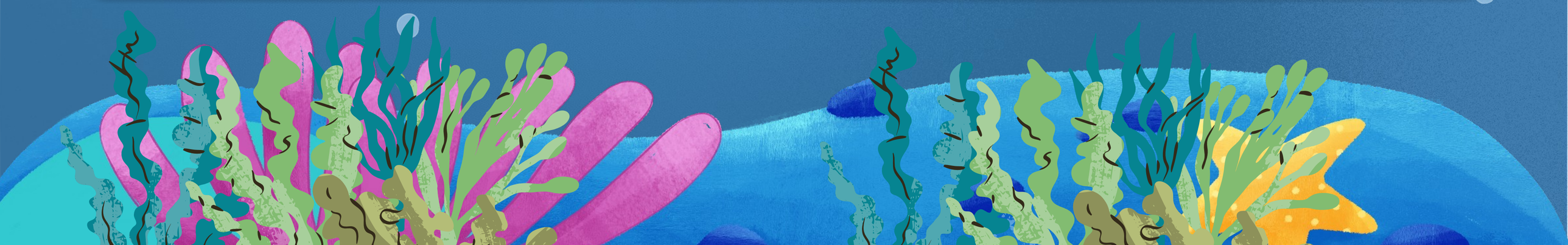
Untuk Pekan 1 sampai 7 dan 13 sampai 15

Essentials of Metaheuristics

Sean Luke, edisi kedua, versi 2.4

Untuk Pekan 9 sampai 12

Unduh dari situs penulis. Tautannya ada di Panduan Peserta.



Penilaian

40%

Latihan

mingguan: kelengkapan dan satu pertanyaan pemahaman

10%

Cek lisan

Pekan 8 individu, Pekan 12 tim

50%

Proyek akhir

tim 30%, cek lisan individu 20%

Kehadiran menjadi syarat: minimal 13 dari 16 pertemuan.

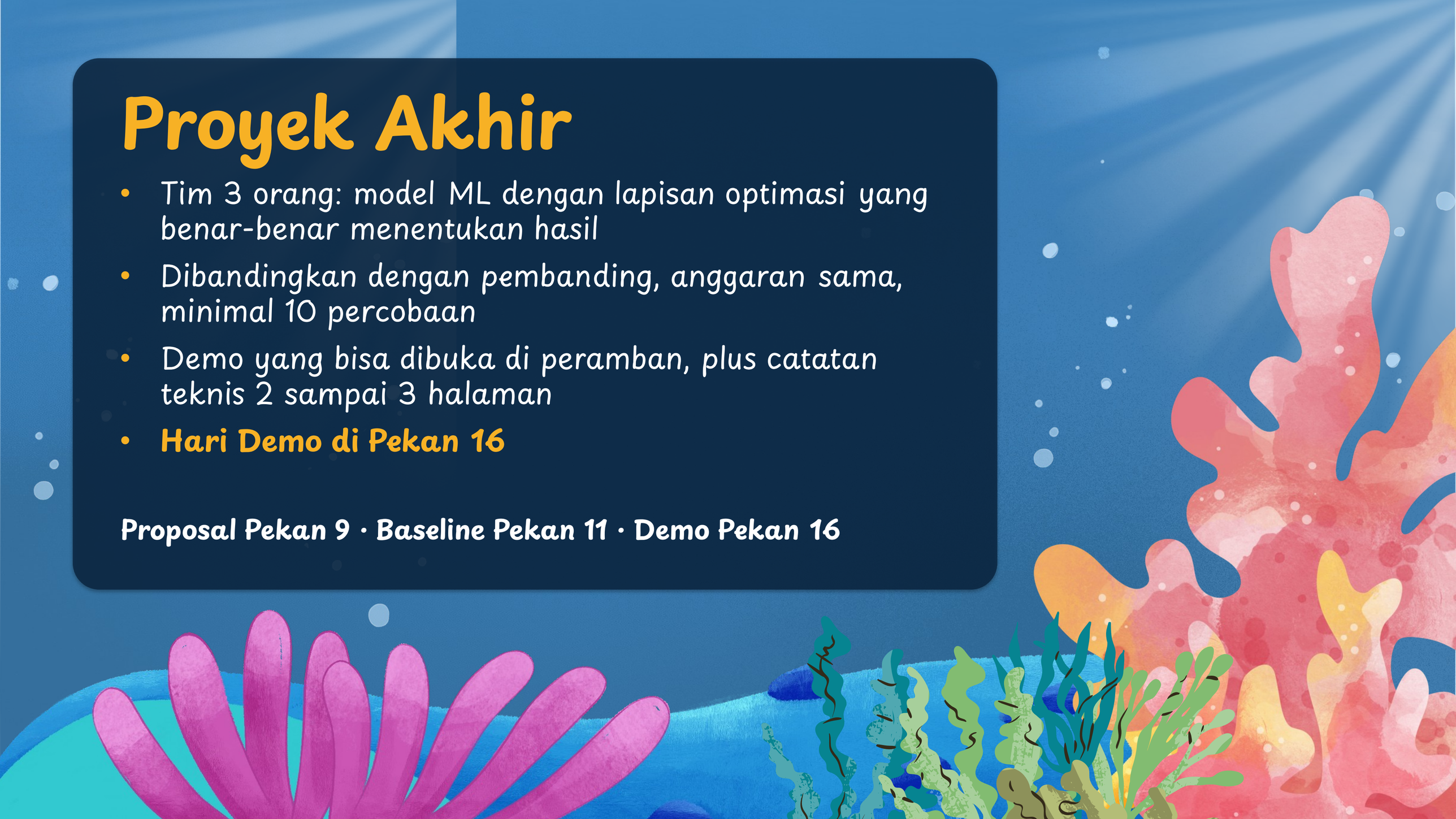




Proyek Akhir

- Tim 3 orang: model ML dengan lapisan optimasi yang benar-benar menentukan hasil
- Dibandingkan dengan pembanding, anggaran sama, minimal 10 percobaan
- Demo yang bisa dibuka di peramban, plus catatan teknis 2 sampai 3 halaman
- **Hari Demo di Pekan 16**

Proposal Pekan 9 · Baseline Pekan 11 · Demo Pekan 16



Alat AI

Pakai AI apa pun, kapan pun. Tidak ada yang mencatatnya.

Tiga hal tetap berlaku:

1. Kuis pembuka dan cek lisan: tanpa perangkat
2. Cek dulu jawaban AI sebelum kamu pakai
3. Kamu yang bertanggung jawab dan harus bisa menjelaskannya

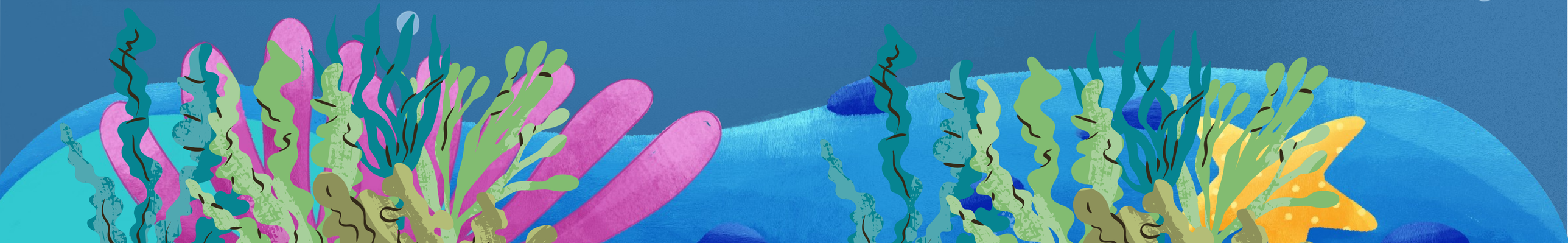
Harapan Dua Arah

Dari kamu

- Hadir dan bawa laptop
- Baca sebelum kelas
- Bertanya saat bingung. Bingung itu normal
- Kumpulkan latihan tepat waktu

Dari mentor

- Panduan baca dan notebook setiap pekan
- Contoh sebelum rumus
- Bantuan saat latihan di kelas
- Umpan balik pada latihanmu



Lalu: persiapan dan game

Istirahat

5 menit



Stasiun Persiapan

A · Google dan Colab

Akun Google masuk. Colab terbuka dan print("halo") berjalan.

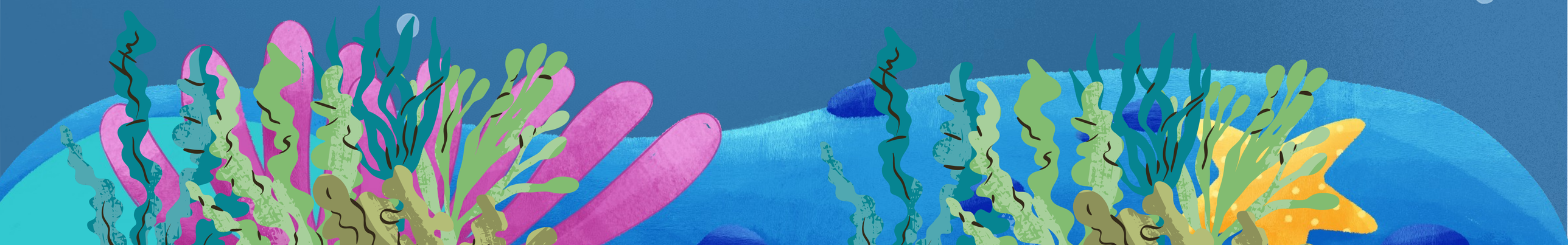
B · GitHub

Akun GitHub dibuat. Username diberikan ke mentor.

C · Buku

Kedua buku terunduh dan terbuka di perangkatmu.

10 menit. Sambil menunggu, buka situs kelas di HP: game berikutnya ada di sana.



Flappy Komodo: Manusia Lawan Evolusi

Babak 1

Kamu bermain

Di HP. Lewati pipa sebanyak mungkin.

10 menit

Babak 2

Evolusi bermain

Di proyektor. 50 komodo belajar sendiri.

10 menit

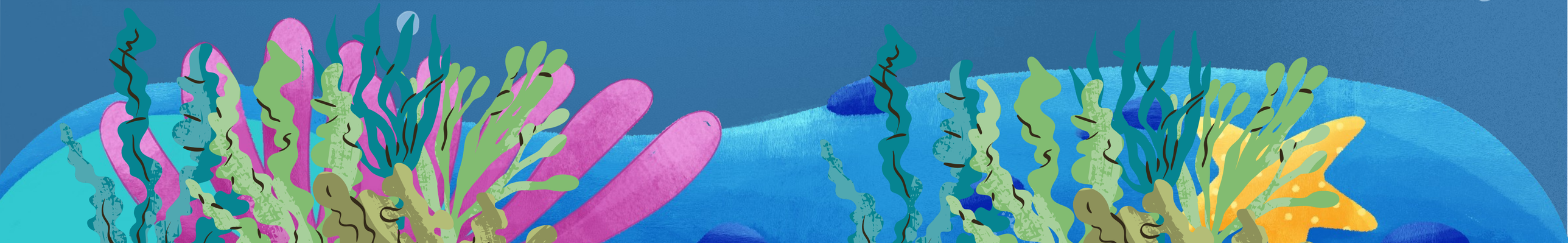
Babak 3

Kamu jadi pelatih

Atur kenop evolusi. Hemat komodo.

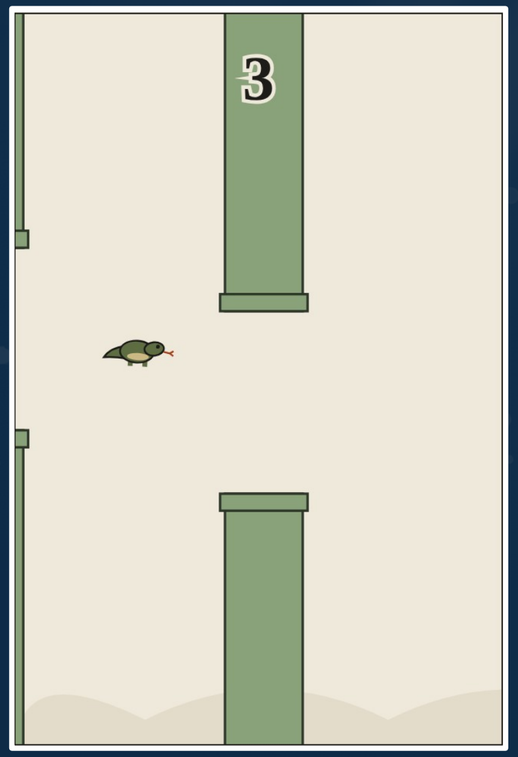
15 menit

Semua orang, dan mesinnya, terbang di lintasan yang sama.



BABAK 1

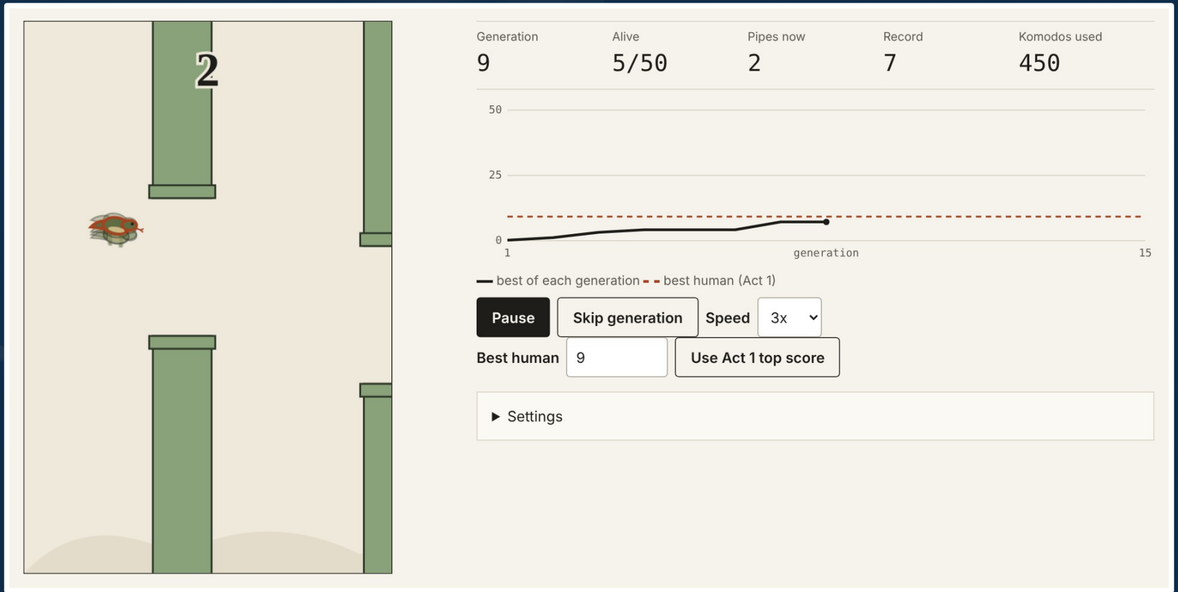
Kamu bermain



1. Buka situs kelas, pilih Flappy Komodo, lalu Play
2. Ketuk layar atau tekan spasi untuk mengepak
3. Main sebanyak yang kamu mau selama 8 menit
4. **Tunjukkan skor terbaikmu ke mentor**

BABAK 2

Evolusi bermain

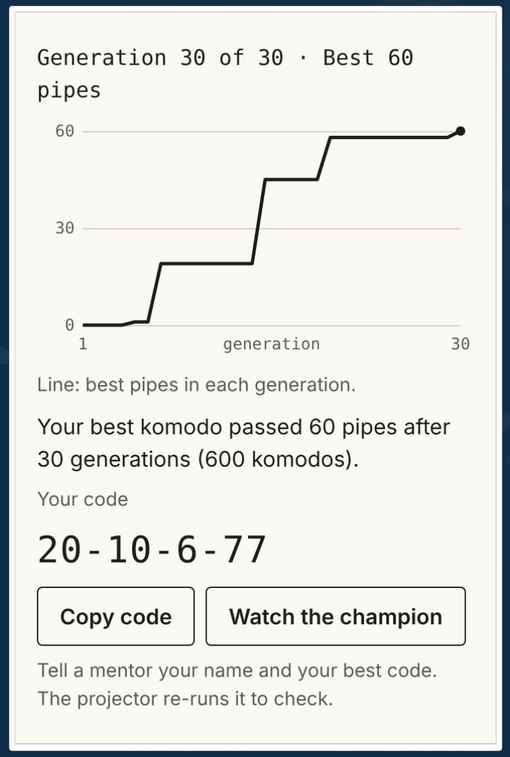


1. 50 komodo, otaknya berisi angka acak
2. Yang terbang paling jauh menjadi induk
3. Anak = campuran induk + sedikit mutasi
4. Ulangi

Tebak dulu: generasi ke berapa mesin melewati skor manusia terbaik?

BABAK 3

Kamu jadi pelatih



- Kenop: populasi, mutasi, ukuran otak, seed
- **Semua dapat anggaran sama: 600 komodo**
- Lintasan makin sempit di setiap pipa
- **Menang: komodo terbaikmu terbang paling jauh**
- Kirim kode terbaikmu ke mentor, contoh 20-10-6-77

Kode yang sama selalu memberi hasil yang sama, jadi proyektor bisa memeriksanya.

PEMBAHASAN

Apa yang barusan terjadi?

Tiga pertanyaan

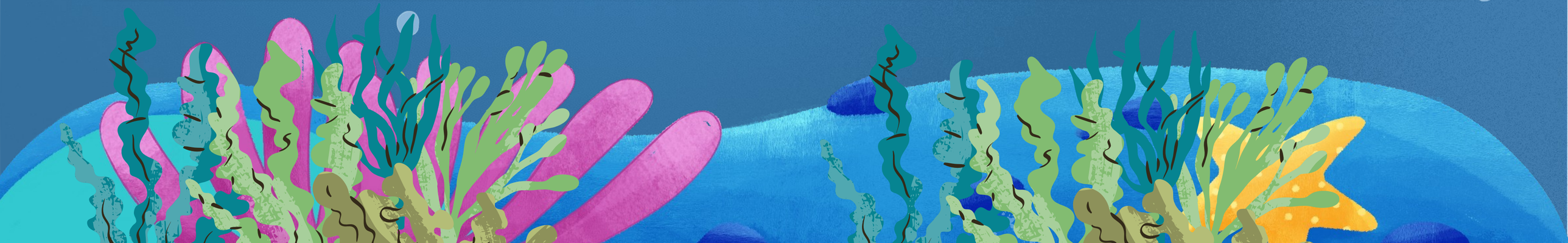
1. Populasi besar berarti generasi lebih sedikit. Mana yang lebih berharga?
2. Seed beruntung atau kenop yang bagus? Bagaimana membedakannya?
3. Tidak ada yang menulis aturan kapan mengepak. Dari mana aturan itu datang?

Otak komodo
jaringan saraf kecil (Pekan 4)

Evolusi
algoritma genetika (Pekan 10)

600 komodo untuk semua
anggaran evaluasi yang sama (Pekan 9)

Cek 5 seed lain
minimal 10 percobaan di proyek akhir



Pemenang dan Pertemuan 1

- Pemenang Babak 1: skor manusia tertinggi
- Pemenang Babak 3: 5 besar dicek dengan 5 seed lain, rata-rata tertinggi menang

Sebelum Pertemuan 1

- Baca UDL bagian 1.1 (halaman 1 sampai 6) dan bab 2 (halaman 17 sampai 22)
- Bawa laptop yang sudah bisa membuka Colab
- Materi pendukung Python untuk programmer Go menyusul di situs kelas

Advance Study Group

Terima Kasih

Sampai jumpa

